

RALPH GUSTAVE

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SUMMARY

Senior Full-Stack Engineer, Research Engineer, and Founding Engineer with 10+ years of experience designing, building and scaling production systems across healthcare, government, defense-adjacent research, communications and e-commerce. Proven contributor to early-stage product research, technical documentation and proof-of-concept development, with hands-on experience supporting SBIR/STTR and government funding submissions. Trusted partner to executive leadership and cross-domain experts, bridging applied research and engineering execution. Expert in React, TypeScript, Node.js, Python/Django, and AWS with deep strengths in cloud-native infrastructure, secure systems and high-availability architectures. Known for translating complex research and technical concepts into actionable systems, proposals, and scalable platforms.

EXPERIENCE

Founding Engineer / Lead Full-Stack Developer / Director of Software Engineering / CIO

Sanctuary Innovations

July 2025 – April 2026, United States, Remote

- Joined as a founding engineer and led the end-to-end design and delivery of a greenfield social media platform from concept to production, establishing the core architecture and engineering roadmap used by the team.
- Developed a full-stack platform using TypeScript, React and Node.js (Express), delivering 30+ REST API endpoints, real-time features with Socket.io and a PostgreSQL database managed via Prisma ORM.
- Built a responsive React frontend with Redux Toolkit and Material UI, delivering a reusable design system documented in Storybook and reducing UI development time by approximately 40%.
- Architected and implemented a secure backend with JWT authentication (httpOnly cookies), RBAC, CSRF protection and rate limiting, resulting in zero critical security incidents in production.
- Designed and deployed AWS infrastructure using Terraform (IaC), provisioning ECS Fargate services, Aurora Serverless v2, Lambda, S3, CloudFront, API Gateway and SQS to support horizontal scaling and high availability.
- Implemented a multi-AZ, auto-scaling architecture with ECS and Aurora, improving system resiliency and supporting 3× traffic growth without service degradation.
- Established CI/CD pipelines with Docker-based builds and automated deployments, reducing deployment time by over 60% and eliminating manual release steps.
- Delivered core social platform features including multi-feed systems, threaded real-time comments, notifications and user profiles, supporting thousands of concurrent WebSocket connections.
- Integrated the AT Protocol (@atproto) to enable decentralized social networking and data portability, expanding platform interoperability and future ecosystem integrations.
- Defined and enforced engineering best practices including TypeScript-first development, ESLint/Prettier standards, structured CloudWatch logging and comprehensive technical documentation to support team onboarding and long-term maintainability.

Lead Engineer

Clawpoint Security Collective

July 2025 – February 2026, United States, Remote

- Consulted with executive leadership and cross-functional teams on early-stage product research and concept development, influencing 2–3 product research tracks and supporting strategic technical decision-making.
- Authored 10+ technical documents, including research briefs, concept papers and multiple proofs of concept, accelerating feasibility validation and reducing early-stage technical risk.
- Prepared and supported 5+ government funding and innovation program submissions, contributing technical narratives, research plans and commercialization sections aligned with agency requirements.
- Contributed to multiple SBIR/STTR proposal efforts, supporting proposals with multi-phase research roadmaps, milestones and technical justifications mapped to DoD and federal evaluation criteria.
- Collaborated with 10+ cross-domain subject-matter experts spanning engineering, defense, applied research and emerging technology domains to refine research scope and technical assumptions.
- Prepared and submitted research materials and proposals to 4+ U.S. government and defense organizations, including DTRA, ECWERX, U.S. Army Corps of Engineers and SOCOM.
- Conducted exploratory and applied research supporting multiple emerging security and defense-focused technology initiatives, informing funding strategy and downstream product development.

Full Stack Developer

iostudio

January 2025 – June 2025, Nashville, TN

- Developed and maintained a government-facing National Guard platform serving thousands of active users, using Laravel and Vue.js to ensure high availability, security and cross-device responsiveness.
- Implemented a hybrid CMS architecture leveraging Drupal and Strapi Router, reducing content publishing time by approximately 35% and improving editorial workflow efficiency.

- Designed and secured RESTful APIs with role-based access control (RBAC), reducing access-related defects and authorization issues by roughly 40%.
- Optimized Vue.js components through lazy loading and performance tuning, improving initial page load times by approximately 30%.
- Built Docker-based CI/CD pipelines that reduced deployment time by over 50% and significantly improved release reliability across staging and production environments.
- Collaborated with designers, product owners, and government stakeholders to deliver multiple mission-critical features on schedule, maintaining 100% on-time delivery under strict government deadlines.

Founding Engineer / Senior Full-Stack Developer

Cerevel therapeutics

February 2021 - November 2024, San Francisco, CA

- Led development of a modern healthcare platform using React and Python/Django, supporting 500+ internal clinical and administrative users across research and operations teams.
- Architected and maintained 20+ scalable REST APIs using Django REST Framework, implementing versioning, filtering, auditing and custom permissions which reduced API-related defects by ~35% and improved long-term maintainability.
- Integrated external healthcare systems using HL7 and FHIR standards, enabling secure exchange of tens of thousands of patient and clinical records while maintaining full auditability and regulatory compliance.
- Designed and implemented RBAC, custom authentication middleware and secure session/token management using Django and Redis, cutting unauthorized access attempts by ~45% and improving security posture.
- Built asynchronous processing pipelines with Celery and RabbitMQ to handle long-running tasks (report generation, bulk data processing), reducing processing time by up to 60% and eliminating UI blocking for end users.
- Optimized PostgreSQL schemas and queries through indexing, query profiling and refactoring, reducing average API response times by ~30% under production load.
- Containerized services with Docker and managed CI/CD pipelines on Google Cloud Platform, reducing deployment time by ~50% and improving release reliability across environments.
- Partnered with product, UX and clinical stakeholders to deliver 10+ major feature releases, improving workflow efficiency and achieving 100% on-time delivery for planned milestones.

Senior Full-Stack Developer

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October 2018 - December 2020, San Jose, CA

- Contributed to the development of a cross-platform video communication product using React (web) and React Native (mobile).
- Integrated WebRTC for real-time video and audio communication, improving call stability and reducing latency by approximately 20%.
- Built and optimized backend services using Node.js and GraphQL, reducing API response times by 25% through resolver and query optimization.
- Designed and optimized MongoDB data models and indexing strategies, improving read performance by 30%.
- Implemented Firebase-based authentication, reducing authentication errors by 40% post-launch.
- Deployed and scaled services on Microsoft Azure, cutting downtime by roughly 50% during peak usage periods.
- Strengthened platform security with end-to-end encryption and RBAC, significantly reducing unauthorized access attempts.

Full-Stack Developer

Amazon

February 2016 - September 2018, Austin, TX

- Contributed to Amazon's e-commerce platform using Next.js, improving page load performance by 40% through server-side rendering and SEO optimization.
- Developed backend services with Laravel, optimizing API performance by 25% for catalog, user and order management systems.
- Built internal tooling and integrations using C#/NET to support automation and operational efficiency.
- Optimized MySQL schemas, indexing and queries, reducing data retrieval times by 30%.
- Led PayPal and Stripe payment gateway integrations, reducing transaction-related issues by 35%.
- Implemented secure JWT-based authentication and AWS Cognito identity management, improving login reliability and user onboarding speed.
- Deployed and optimized infrastructure on AWS to support high-traffic events while minimizing downtime.

EDUCATION

Bachelor of Science in Computer Science

Florida State University • Tallahassee, FL • 08/2011-12/2015

CERTIFICATIONS

AWS Certified Cloud Practitioner

CLF-C02 • 2024

SKILLS

Languages: TypeScript, JavaScript, Python, PHP, C#, HTML, CSS

Frontend: React, Next.js, Angular, Vue.js, Redux Toolkit, RxJS, NgRx, Material UI, Tailwind CSS, SCSS, Storybook

Backend: Node.js (Express, NestJS), Django, Django REST Framework, FastAPI, Flask, Laravel, GraphQL, ASP.NET Core

Real-Time & Messaging: Socket.io, WebSockets, SQS, Redis (ElastiCache)

Databases & Data: PostgreSQL, MySQL, MongoDB, AWS DynamoDB, Prisma ORM, RDS Proxy

Cloud & DevOps: AWS (ECS Fargate, Aurora Serverless v2, Lambda, S3, CloudFront, API Gateway), Google Cloud Platform, Azure, Docker, Kubernetes, Terraform (IaC), CI/CD Pipelines

Security & Auth: JWT (httpOnly cookies), Role-Based Access Control (RBAC), CSRF Protection, Rate Limiting, OAuth, Security Headers (Helmet.js)

Testing & Quality: Jest, Cypress, PyTest, ESLint, Prettier

Monitoring & Logging: AWS CloudWatch, Structured Logging

Tools & Collaboration: GitHub, GitLab, SVN, Jira, Trello, Slack
